Dandrell E. Johnson II

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Mechanical Engineering Major

I'm an ambitious student seeking an internship to apply my skills and make meaningful contributions to a dynamic team, take ownership, and show my technical proficiency.

SKILLS

LABORATORY: Programming Microcontrollers • Rapid prototyping • Precision Machining
SOFTWARE: C • C++ • C# • MATLAB • Java • Python(Pytorch • Colab Notebook)
CAD/CAM: Solid Edge • SolidWorks • AutoCAD • Autodesk Inventor • Mastercam • Cura

EDUCATION

Alabama A&M University- Normal, AL — May 2027
Bachelor of Science in Mechanical Engineering (BSME) GPA: 3.0/4.0
- Consultation in Manufacturing Engineering

Minor of Science in Computer Science (MSCS) GPA: 3.0/4.0

University of Alabama at Birmingham - Birmingham, AL — August 2018
School of Engineering **Received Certificate for Automated & Robotics**

WORK HISTORY

Heil Environmental Fort Payne, AL — August 2025 – Present
Manufacturing Engineering Intern
- Conducted time studies and Standardized 20+ workflow analyses to optimize assembly operations, reducing cycle times by 10% and identifying bottlenecks for the process.
- Integrated training videos and digital documentation into manufacturing systems (Smartsheet/L2L), enhancing operator onboarding and reducing operational errors by 25%
- I designed and deployed a live time-study tool that cut engineering effort ~52% and got adopted on the floor

Boys and Girls Club of North Alabama Huntsville, AL — August 2024 – Present
STEM Intern - (Robotic Coach)
- Provided a team of ten students with hands-on learning experience involving critical thinking, problem-solving skills, and gaining interest in the vast STEM fields.
- Challenged children to learn communication skills and conduct research effectively by involving them in The First Robotics LEGO® League.

Commercial HVAC Trane Technologies La Crosse, WI — January 2025 – August 2025
Project Engineering Intern
- Led the installation of a new assembly cell as part of a new production line, which increased revenue by $1.3 million and plant capacity by 30%
- Developed new standard work for operations by calculating cycle time and performing time studies. Saved over $20,000 in materials by understanding facility capacity with customer demand to eliminate the holding cost of product.

National Center for Supercomputing Applications (NCSA) Urbana/Champaign, IL — May 2024 – July 2024
Research Intern
- Develop a method for analyzing cracks in concrete that correlates them with applied loads.
- Utilized PyTorch and Colab for different segmentation models to run datasets over 10,000 images.

Alabama A&M Special Projects Laboratory Normal, AL — October 2022 – July 2023
FORMULA SAE - (Project Supervisor)
- Led the first HBCU team in Formula SAE to build a competition-ready race car.
- Integrated Arduino controllers to implement paddle shifters.
- Conducted flow simulations on the intake of the vehicle systems, providing data on the emissions standard, and for the highest performance output.

NASA Rover Project - (Test and Evaluation Engineer)
- Analyzed the rover chassis and its components to verify that it will be able to handle various terrains.
- Presented the full research results to NASA.

Memberships & Certification

- National Society of Leadership and Success (NSLS)
- AAMU Engineering Dean's List 2023
- PLTW Computer Science and Engineering Academy
- Society of Automotive Engineers (SAE) Member
- National Society of Black Engineers (NSBE)
- AAMU Marching Maroon and White Band 2023
- Revit Autodesk Certified User
- Inventor Autodesk Certified User
- Certificate in Automation & Robotics
- Certificate in Trojan Safe Computers